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Week 14 Quiz

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1. Moisture content is a major factor affecting radar scattering from surfaces because (choose one):

Due Sep 22, 11:59 PM IST Attempts 3 every 2 hours

1 / 1 point

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- ☐ It increasingly makes the surface look specular
- ☒ It affects the surface dielectric constant
- ☐ evaporating moisture reflects the incident radar beam
- ☒ Correct That is correct.

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2. Sub-surface radar imaging is possible if (choose one):

1 / 1 point

- ☐ short wavelengths are used
- ☒ long wavelengths are used and the material is dry
- ☐ medium wavelengths are used and the surface-air interface is very smooth
- ☒ Correct That is correct. It is under those conditions that best penetration is possible.

3. In radar imaging the Rayleigh criterion is useful for assessing whether (choose one):

1 / 1 point

- ☒ a surface is likely to behave as smooth
- ☐ volume scattering will dominate over surface scattering
- ☐ surface scattering will dominate over volume scattering
- ☒ Correct That is correct.

4. Which of the following scattering mechanisms has the weakest dependence on incidence angle?

1 / 1 point

- ☐ smooth surface scattering
- ☐ rough surface scattering
- ☒ volume scattering
- ☒ Correct That is correct.

5. When imaging a forest with SAR, if you wanted the response to be dominated by the canopy, with little or no contribution from the trunks or ground what wavelength would you use?

1 / 1 point

- ☒ X band
- ☐ C band
- ☐ P band
- ☒ Correct That is correct. It will give maximum return from the canopy with little penetration to the woody matter or ground

6. Which of the following mechanisms is regard as a strong scatterer in remote sensing radar imaging?

1 / 1 point

- ☐ a rough surface
- ☒ a dihedral corner reflector
- ☐ a wet, smooth surface
- ☒ Correct That is correct. A corner reflector is a classic strong scatterer.

7. The radar response of a forest at long wavelengths will be strongly affected by the surface on which the forest sits if (choose one):

1 / 1 point

- ☐ the forest floor is covered by a volume scattering understory
- ☒ the forest floor is a strong specular reflector
- ☐ the forest floor is very rough
- ☒ Correct That is correct.

8. Sea surface features are best detected with SAR (choose one):

1 / 1 point

- ☒ at a low incidence angle with C or L band imagery
- ☐ at low incidence angle with X band imagery
- ☐ at high incidence angle with P band imagery
- ☒ Correct That is correct.

9. The cardinal effect is caused by (choose one):

1 / 1 point

- ☒ changing alignments of rows of strong scatterers with respect to the incoming radar beam
- ☐ wind rippling of the surface of sea water
- ☐ strong scatterers underneath tree canopies
- ☒ Correct That is correct.

10. Oil slicks can be detected in SAR imagery because (choose one):

1 / 1 point

- ☐ oil has a high dielectric constant
- ☐ oil is a volume scatterer whereas the ocean surface is a Bragg scatterer
- ☒ Oil dampens the capillary waves that the radar energy couple into
- ☒ Correct That is correct